

LARGE ANIMAL PRACTICE- REFRESH

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LEARNING IS A CONTINUOUS PROCESS



ANTIBIOTICS

¢ OTC

¢ Strepto Penicillin

¢ Enrofloxacin

¢ Gentamicin

¢ Sulpha Trimethoprim



- ¢ Amoxycillin plus Cloxacillin
- ¢ Ceftriaxone
- ¢ Cefotiofur sodium
- ¢ Tylosin
- ¢ Tilmococsin
- ¢ Orbifloxacin
- ¢ Marbofloxacin



PURGATIVES IN CATTLE

¢ MAGNESIUM SULPHATE 500 GM

SODIUM CHLORIDE - 250 GM

TRADE NAMES

BOVILAX

LUXABULK



FLUIDS

¢ NORMAL SALINE

¢ DNS

¢ RINGER LACTATE

¢ DEXTROSE 20,50



NSAID

- ¢ Melonex
- ¢ Flunixin meglumine
- ¢ Phenybutazone
- ¢ Ketoprofen
- ¢ Tolfenamic acid



STYPTICS

¢ Sytptochrome

¢ Menadione sodium

¢ Botropase



CORTICOSTEROIDS

¢ DEXAMETHASONE

¢ PREDNISOLONE

¢ METHYL PREDNISOLONE

¢ ISOFLUPRIDONE



DISEASES OF CALVES

- ¢ HYPO VITAMINOSIS A
- ¢ CALF SCOUR
- ¢ SALMONELLOSIS
- ¢ WORM INFESTATION
- ¢ CALF PNEUMONIA
- ¢ NAVEL ILL
- ¢ MILK OVER LOAD OR RUMINAL DRINKERS
- ¢ REGURGITATION
- ¢ BUFFALO CALF – TOXOCARA????
- ¢ RULE OUT CONGENITAL ANOMALIES



COLOSTRUM FEEDING

- ¢ 10 % of Body weight within few hours of birth
- ¢ Field test to rule out colostrum deprived animals



DEWORMING OF CALVES

¢ Buffalo calf – Day old

¢ Pyrantel pamoate

¢ Piperazine adipate powder

¢ Albendazole

¢ Fenbendazole



HYPOVITAMINOSIS A

¢ Incoordination

¢ Rotation of head

¢ Ataxia

¢ Treatment : Vit A 440 iu/kg IM



CALF SCOUR (COLLIBACILLOSIS)

- ¢ White colour diarrhea
- ¢ Dehydration
- ¢ E.coli infection
- ¢ Treatment : Fluid Therapy
Antibiotics



SALMONELLOSIS

¢ Diarrhea with mucus and blood

¢ Straining

¢ Dehydration

¢ Treatment : Fluid therapy
Antibiotics



CALF PNEUMONIA

- ¢ Pyrexia
- ¢ Respiratory distress
- ¢ Dyspnea
- ¢ Oral mouth breathing
- ¢ Lungs- Exaggerated sounds

¢ Rx : Antibiotics

Steroids

Bronchodilators (Deriphylin Inj)

Frusemide

No Fluids



JOINT ILL / NAVEL ILL

- ¢ Swelling of the joints
- ¢ Inappetance to anorexia
- ¢ Abscess – pain on palpation of joints
- ¢ Respiratory distress

- ¢ Rx: Antibiotics
 - Steroids
 - Surgical drainage of wound
 - Povidone seton
 - Magsulph glycerin seton



SKIN LESIONS IN CALF

¢ Erythematous

¢ Alopecia

¢ No Itching

¢ Normal moulting

¢ Rx: Symptomatic – Vit A and supplements



HYPOMAGNESAEMIA TETANY - CALF

- ¢ Frenzy
- ¢ Excited
- ¢ Salivation
- ¢ Animals fed with only milk
- ¢ No roughages
- ¢ Body weight more than 50 Kg

- ¢ Rx: Magnesium solutions
- ¢ 10% Mag sulphate



CUTANEOUS MYIASIS

¢ Maggot wounds

¢ Predilection sites- Hoof,,umblicus,,, vulva,, etc...

¢ Rx: Acaricides

Negashunt powder

Turpentine oil

Eucalyptus oil





OTHER CONDITIONS –CALF & HEIFERS

- ¢ Blood protozoan diseases

- ¢ Theileriosis

- ¢ Babesiosis

- ¢ RECUMBENCY IN HEIFERS



CONGENITAL

- ¢ Cleft palate
- ¢ Contracted tendon
- ¢ Mega esophagus
- ¢ Atresia ani
- ¢ Atresia recti



ECTOPARASITE CONTROL

¢ Ivermectin

¢ Cypermethrin Wash

¢ Pour on preparations



UDDER DISEASES

¢ Physiological Udder edema

¢ Cow Pox, Pseudo Cow Pox

¢ Bovine Ulcerative mamilitis

¢ Udder Impetigo

¢ Udder wart

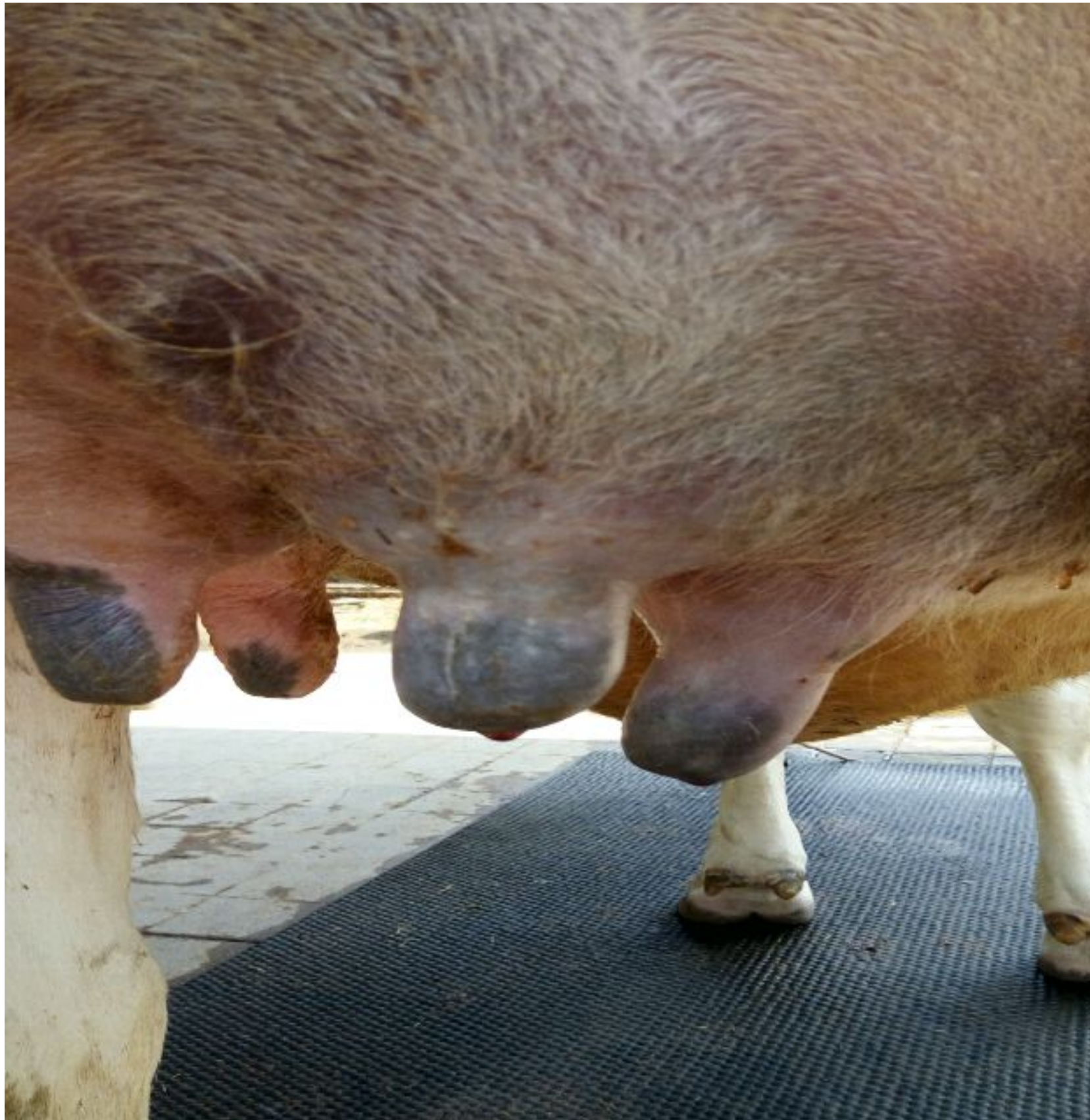
¢ Intertrigo



PHYSIOLOGICAL UDDER EDEMA









HAEMGALACTIA – ROSE MILK

¢ Inj.Styptics

¢ Inj.Calcium borogluconate

¢ Oral styptics



SUB CLINICAL MASTITIS -CMT TEST











DO'S IN MASTITIS

- ¢ Early Attention
- ¢ Correct Dosage
- ¢ Selection of Antibiotics according to previous experience
- ¢ Correct duration (Minimum of 5 to 7 day)
- ¢ Frequent stripping of Milk
- ¢ Supportive Treatments
- ¢ Vit E and Anti inflammatory
- ¢ External applications
- ¢ Teat Dipping in recurrent mastitis



DON'T IN MASTITIS

¢ Delayed Cases

¢ Old cases

¢ Unsterile needle intramammary infusions

¢ Expired intra mammary infusions

¢ Unhygienic teat infusions



SEQUELAE TO MASTITIS

- ¢ Toxemia
- ¢ Corneal opacity
- ¢ Pneumonia
- ¢ Polyarthrititis
- ¢ Fibrosis of udder
- ¢ Teat obstruction
- ¢ Recumbency
- ¢ Death



TEAT DIPPING

- ¢ Povidone iodine

- ¢ Khorsoin solutions

- ¢ Teat sprays

- ¢ Many trade names available



PREVENTION OF RECURRENT MASTITIS

- ¢ Check the cattle shed (concrete or muddy)
- ¢ Environment hygiene
- ¢ Cleanliness
- ¢ Feeding immediately after milking
- ¢ Don't make animal to lie down immediately after milking



¢ Milk let down -- Increlac bolus

Oxytocin

¢ Leaky teats – Phosphorus supplement

Reduce the interval of milking

¢ Hard milkers- Heifers ··??

Fomentation

Calcium administration

¢ Agalctia

Calcium administration

Metoclopramide

Protein feed supplements



DRY COW THERAPY

- ¢ Slowly Tapper Milking
- ¢ Antibiotics -Intramammary



COW POX, PSEUDO COW POX



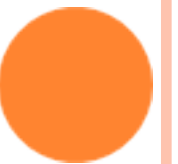
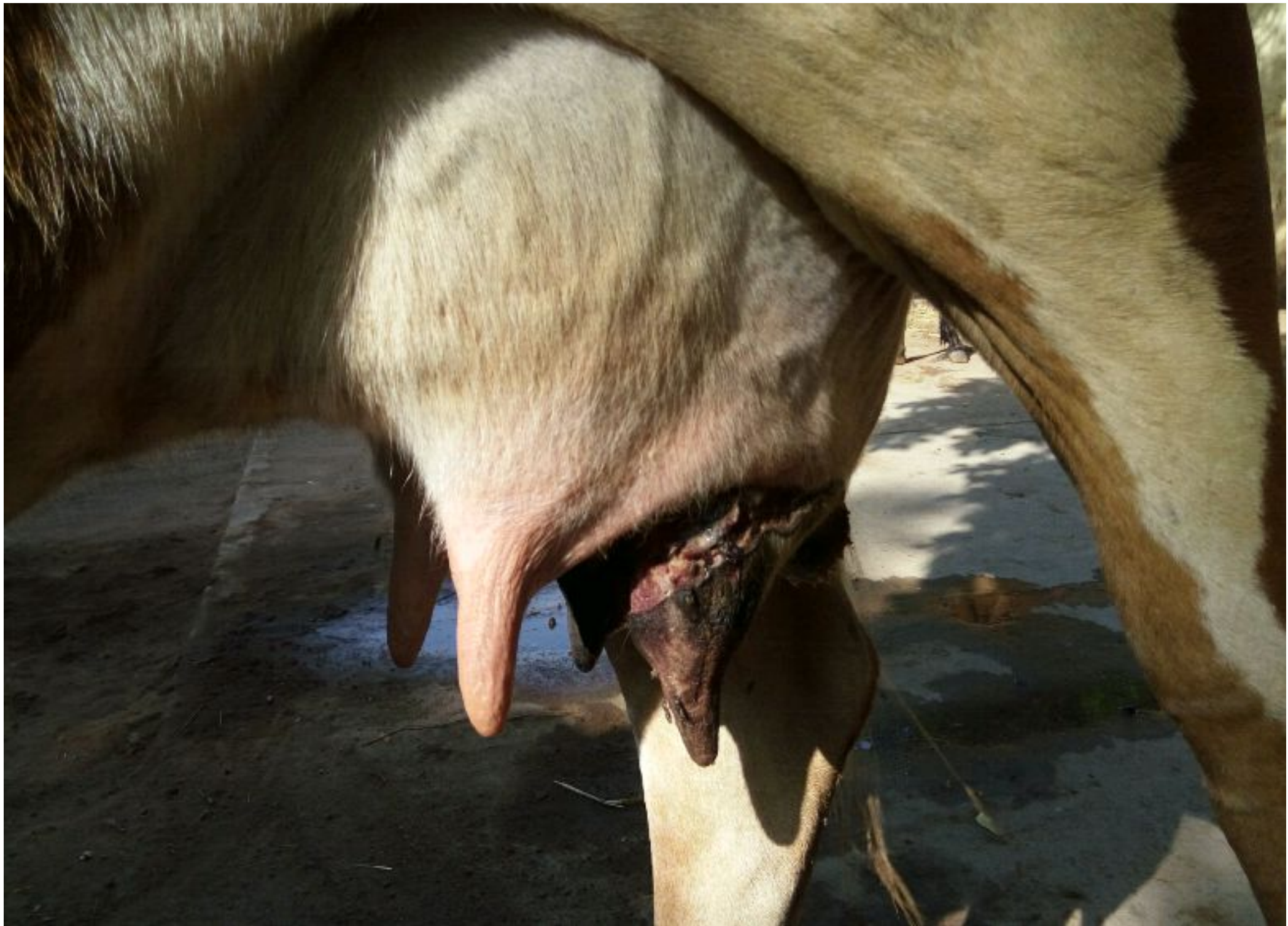
BOVINE ULCERATIVE MAMMILITS



WART







INTERTRIGO

¢ Inj. Antibiotics

¢ Inj. Anti inflammatory



ETHNO VET PREPARATION

¢ Aloe vera (kathalai)

¢ Turmeric (Manjal)

¢ Calcium carbonate (Sunnambu)



Temperature



- ¢ Viral diseases
- ¢ Blood protozoan diseases
- ¢ Inflammatory Causes
- ¢ Bacterial Diseases
- ¢ Toxemia or septicemia



VIRAL

¢ E fever

¢ FMD

¢ MCF

¢ IBRT



EPHEMERAL FEVER



- ¢ Lameness
- ¢ Nasal discharge
- ¢ Reduced dung passage/ hard or pellety dung
- ¢ Temperature
- ¢ Lymph node enlargement
- ¢ CMM- Congested or pink

- ¢ Rx: Anti Inflammatory
Antibiotics



COMPLICATIONS OF EF

¢ Recumbency

¢ Treatment with calcium
advisable



MCF

¢ Common in Buffaloes

¢ Cow also

¢ Respiratory form

¢ Alimentary form









Protozoan diseases

¢ Anaplasmosis

¢ Theileriosis

¢ Babesiosis

¢ Trypanosomiasis



Anaplasmosis

Inj.OTC 22 mg/kg IV x 3 days

Blood transfusion if Pcv below 20%











Theileriosis

- ¢ Inj. Buparvoquone 2.5 mg/kg IM
- ¢ Blood Transfusion if Pcv below 20%







Babesiosis

¢ Inj. Berinil 3.5 mg/kg IM

¢ Blood Transfusion if pcv < 20%





Trypanasomiasis

¢ Inj. Berenil IM

¢ Inj. Quanapyramine Sulphate

¢ Inj. Quanapyramine Chloride
2.5gm SC



TRYPS...:

¢ Wet film examination

¢ Blood smear

¢ Limb edema?

¢ Hypoglycemia?

¢ Nervous signs

¢ Anemia

¢ Corneal opacity?



BACTERIAL

¢ Hemorrhagic septicemia

¢ Black Quarter

¢ Tetanus

¢ Botulism



HEMARAGHIC SEPTICEMIA

- ¢ Pyrexia
- ¢ Jowl edema
- ¢ Respiratory distress
- ¢ Dyspnea
- ¢ Petichation cmm
- ¢ Exaggerated lung sounds
- ¢ Rx:
- ¢ Enrofloxacin,
- ¢ Suplhatrimethoprim,
- ¢ OTC,
- ¢ Ceftriaxone



BLACK QUARTER

- ¢ Clostridium chauvoei
- ¢ Pyrexia
- ¢ Swelling of thigh muscles
- ¢ Gluteal & Lumbar muscles
- ¢ Crepitation sound
- ¢ Serosanguinous fluid from the affected area

Rx:

- ¢ Penicillin 40000 iu per kg



BQ



ACTINOMYCOSIS





ACTINOMYCOSIS

¢ Inj. Streptopenicillin 5 gm IM

¢ Potassium iodide 10gm orally daily for 10 days









TETANUS

¢ Rx:

¢ Large dose of Penicillin

¢ Muscle relaxants

¢ Metronidazole

¢ Prognosis: Grave in cattle





PINK EYE

¢ Common is Sheep and Goats

¢ Cattle

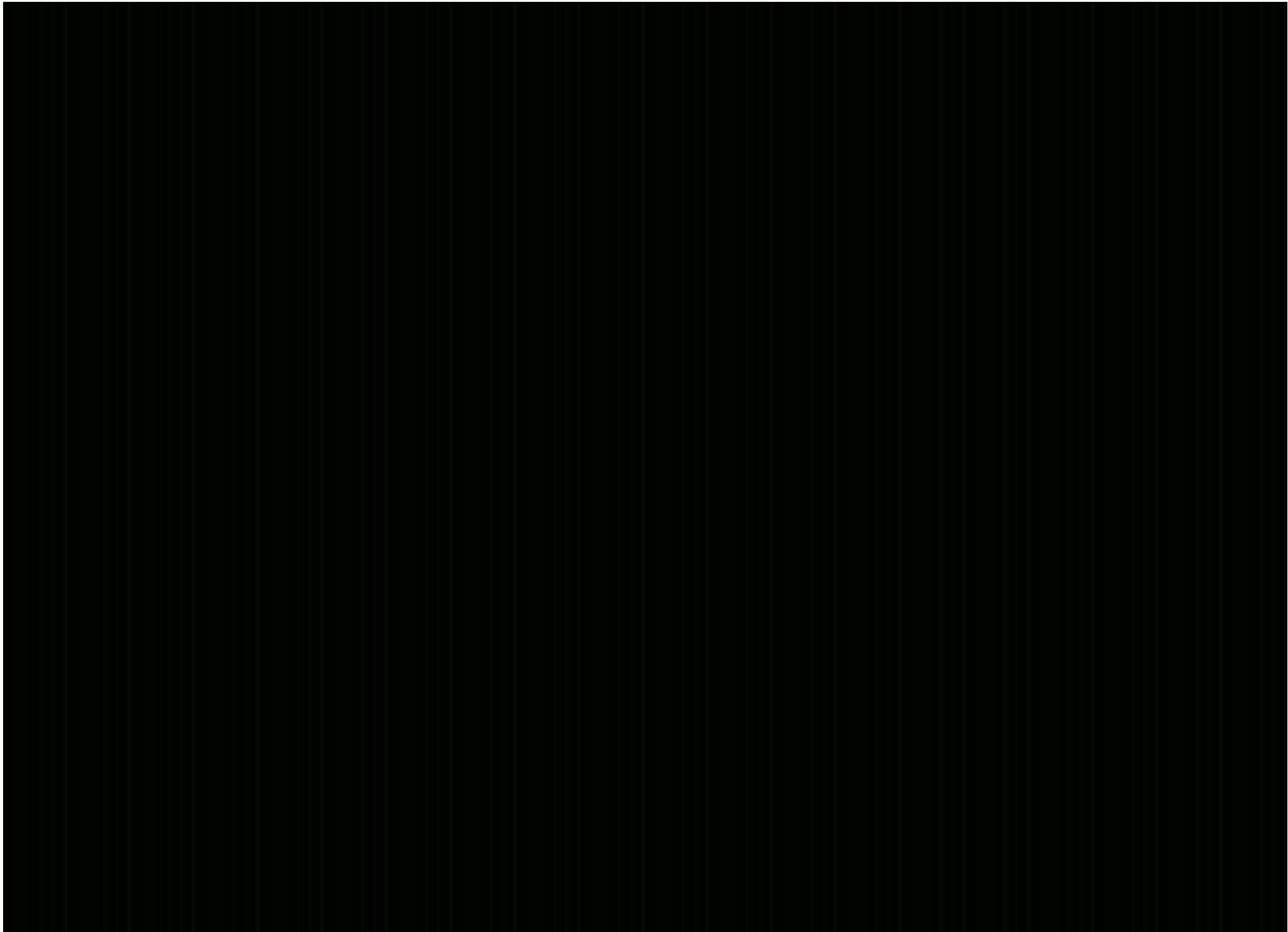
¢ Treatment

¢ Antibiotics

¢ Ophthalmic ointments







LINE OF RX – ASPIRATION PNEUMONIA

¢ Enro, Ceftiofur, SP

¢ Steroids

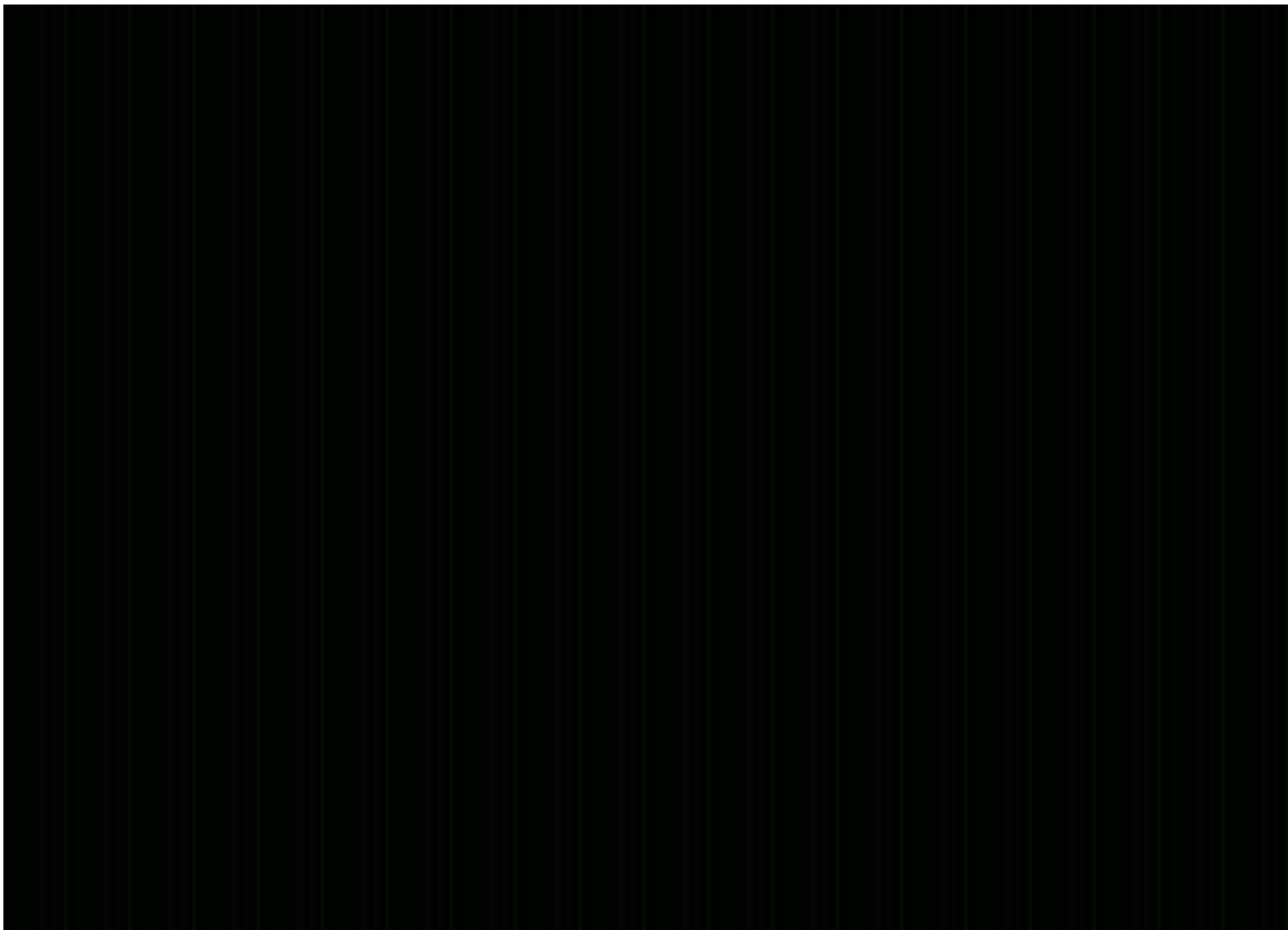
¢ Bronchodilators (Deriphyllin)

¢ No Fluids

¢ Lasix

¢ Antihistamines







METABOLIC DISEASES

Milk Fever

¢ Ketosis

¢ Hypomagnesaemia

¢ Downer Cow

¢ Low Milk fat Syndrome

¢ Post Parturient Haematuria



MILK FEVER

¢

¢ INJ.CALCIUM BOROGLUCONATE

¢ Half dose IV and remaining SC for better result



KETOSIS

- ¢ Primary- metabolic cause
- ¢ Secondary- due to any other diseases



¢ Wasting form

¢ Nervous form

¢ Alimentary form

¢ Starvation ketosis



- ¢ Due to negative energy balance
- ¢ Improper ration
- ¢ Low feed supply in lactation



KETOSIS -WASTING FORM

- ¢ Inappetance
- ¢ Reduction in milk yield
- ¢ Woody appearance
- ¢ Gait abnormality
- ¢ Staggering gait
- ¢ Vision impairment
- ¢ Vital signs-Normal



NERVOUS FORM

- ¢ Convulsions
- ¢ Self biting
- ¢ Circling
- ¢ Apparent blindness
- ¢ Aimless walking
- ¢ Wandering
- ¢ Reluctant to move
- ¢ Gait abnormality



- ¢ Falling down on touching animal
- ¢ Circling
- ¢ Chewing movement with salivation
- ¢ Head pushing
- ¢ Vigorous licking of skin



CLINICAL PATHOLOGY

¢ Blood glucose

¢ Ketone bodies in urine,milk



DIFF DIAGNOSIS

¢ Rabies

¢ Hypomagnesaemic tetany



DIAGNOSIS

¢ Clinical signs

¢ Rothra s' test- milk and urine

¢ B hydroxybutyrate strip test

¢ Response to treatment



TREATMENT

¢ Inj. 20 % dextrose 1 – 1.5 litres IV

¢ or

¢ Inj. 50% dextrose IV

¢ Inj. Dexamethasone 40 mg IM

¢ Inj. Insulin 200-300 iu SC 24 hr interval



¢ Propylene glycol or Glycerine 220 gm PO

¢ Sodium lactate or ammonium lactate PO

¢ Cobalt and vit B12 supplemetation









PREGNANCY TOXEMIA- SHEEP

- ¢ In twins or triplets
- ¢ Separation from flock
- ¢ Apparent blindness
- ¢ Ophisthotonus
- ¢ Stand still
- ¢ Reluctant to move
- ¢ Dry scant feces
- ¢ Muscular tremors
- ¢



CONTD...

¢ Convulsions

¢ Star gazing appearance

¢ Recumbancy

¢ Depression



¢ Cold or inclement weather

¢ Negative energy balance

¢ Stress



CLINICAL PATHOLOGY

¢ Low blood glucose- not always??

¢ Elevation of plasma cortisol level



TREATMENT

¢ Glucose injections

¢ Insulin 20-40 iu - alternate days IM

¢ Electrolyte fluids

¢ Induction of parturition



TAKE HOME POINT

- ¢ No response to treatment when sheep becomes recumbent
- ¢ No response to 50% dextrose in recumbent sheep
- ¢ Provide electrolyte fluids



DIFFERENTIAL DIAGNOSIS

¢ Polioencephalomalacia



LOW MILK FAT SYNDROME

- ¢ Vinegar orally
- ¢ Split feeding of Carbohydrates'
- ¢ More Roughages
- ¢ Magnesium oxide orally



Post Parturient Hbnuria

¢ Sodium Acid Phosphate 60gm iv and
60 gm sc

*Blood Transfusion if pcv < 20%





GI DISORDERS

- ¢ SIMPLE INDIGESTION
- ¢ ACID INDIGESTION
- ¢ VAGAL INDIGESTION
- ¢ BLOAT – FREE AND FROTHY
- ¢ RUMEN IMPACTION
- ¢ ABOMASAL ULCERS
- ¢ PARALYTIC ILEUS
- ¢ INTESTINAL OBSTRUCTION
- ¢ INTESTINAL INTUSSUSCEPTION



RUMINAL ACIDOSIS

¢ Mild

¢ Moderate

¢ Severe

¢ SARA



DIAGNOSIS

- ¢ Rumen fluid examination
 - Protozoa density- grading



TREATMENT- RUMINAL ACIDOSIS

- ¢ Evacuate The Rumen Contents
- ¢ Inj.Sodium Bicarbonate
- ¢ Inj.Normal Saline
- ¢ Antihistamines
- ¢ B.Complex
- ¢ Oral antibiotics
- ¢ Oral Antacids
- ¢ Probiotics
- ¢ Cud transplantations



RUMINAL ACIDOCIS CALCULATION

$0.3 \times \text{BASE DEFICIT} \times \text{Body wt} = \text{m.eq of}$
 NaHco_3

¢ 1ml = 0.9 m.eq (or) 1 m.eq of NaHco_3

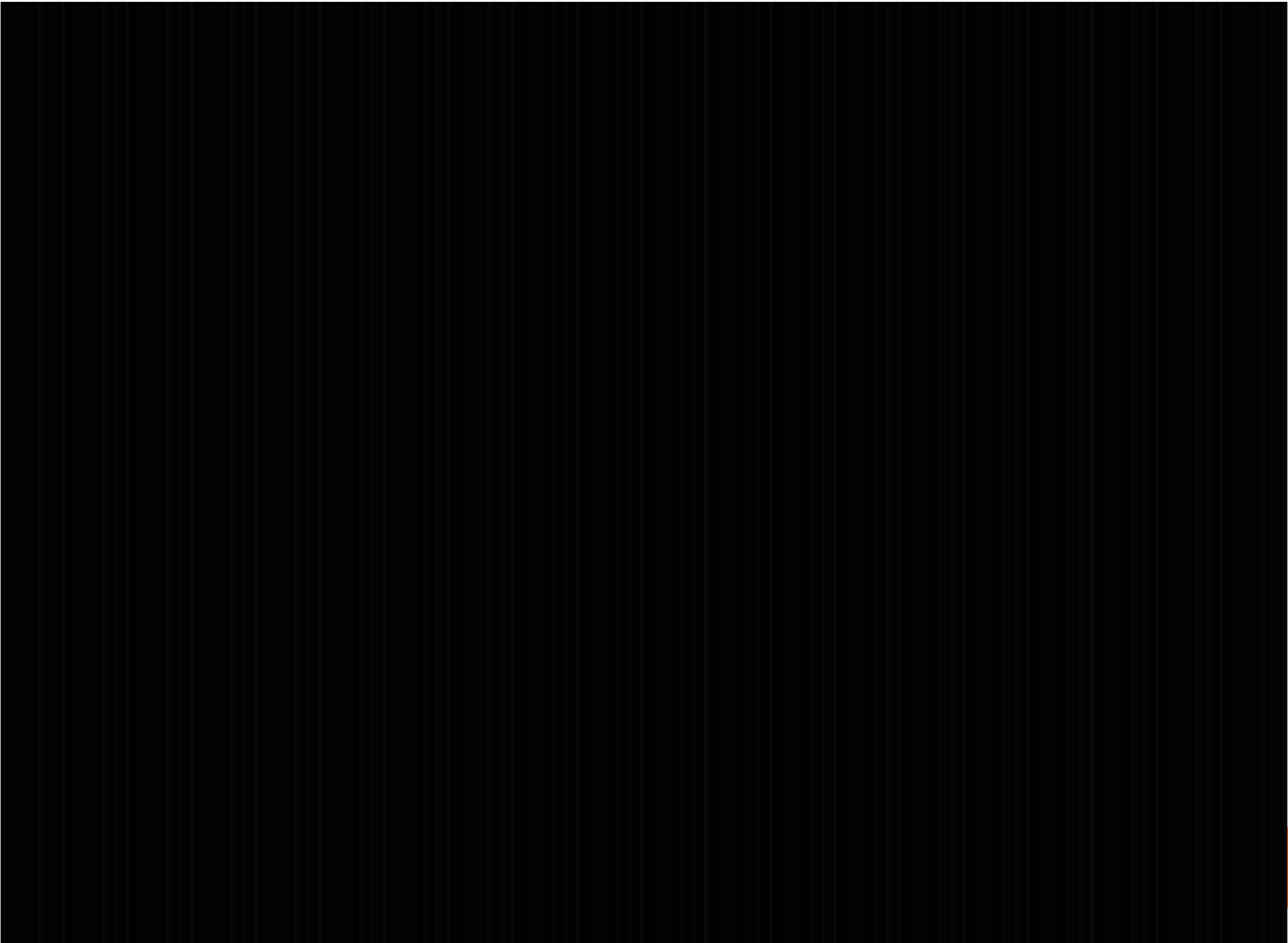


Thump Rule.....

¢ 1 ml / kg Mild to moderate

¢ 2-3 ml / kg Severe Ruminant Acidosis





FREE GAS BLOAT - CHOKE

- ¢ Salivation
- ¢ Extended neck
- ¢ Frequent jerking cough
- ¢ Tympany
- ¢ Respiratory distress



FROTHY BLOAT

- ¢ Liquid paraffin 500 ml to 1 litre
- ¢ Antacid with silicon dimethicone 500 ml
- ¢ Vegetable oils
- ¢ Fluids IV



VAGAL INDIGESTION

¢ INAPPETANCE

¢ BRADYCARDIA

¢ HYPERMOTILITY OF RUMEN

¢ PALE SHAPED ABDOMEN

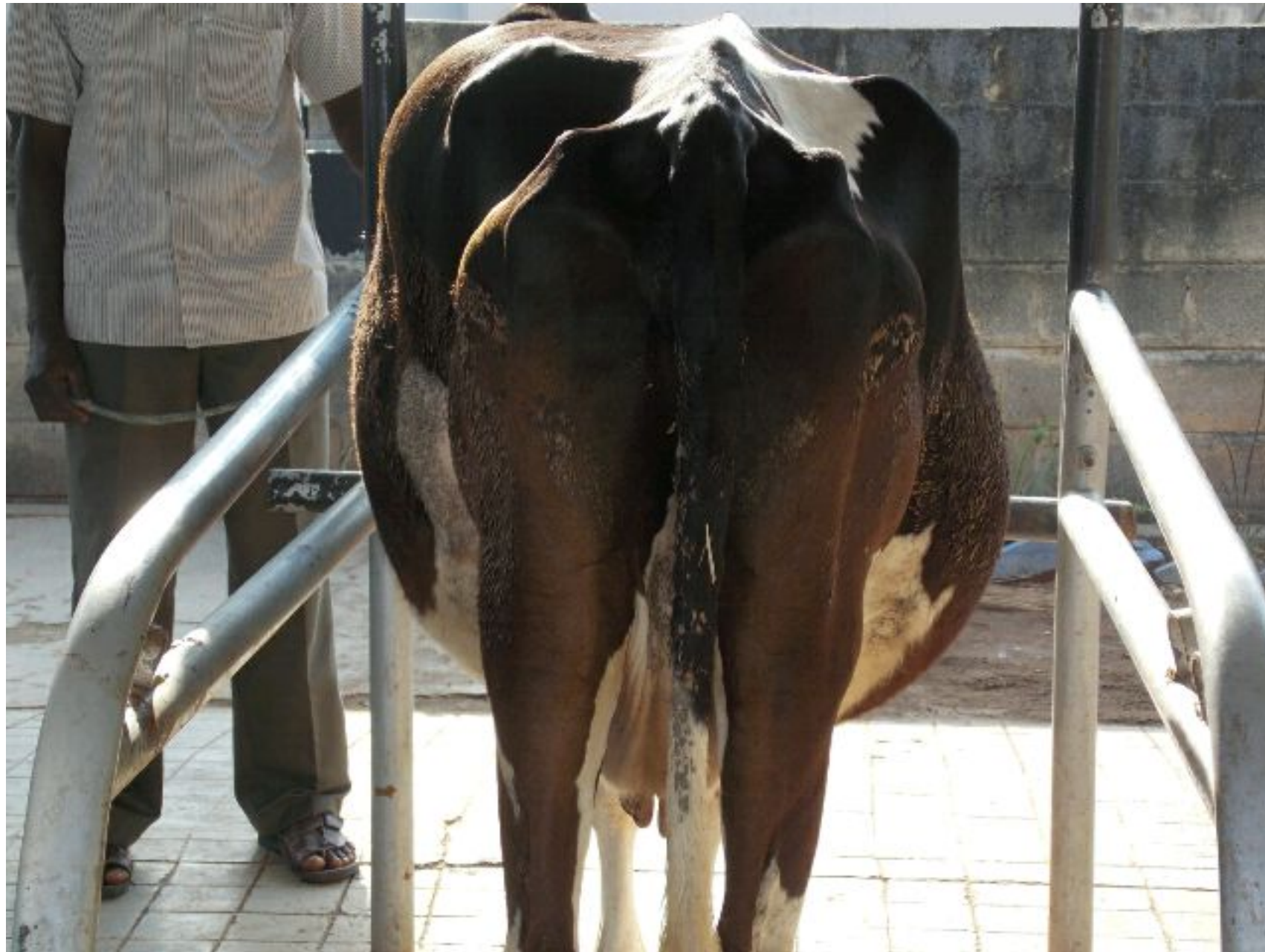
¢ SCANTY FECES







VAGAL INDIGESTION





Not Passing Dung....??

- ¢ Rumen Impaction
- ¢ Intestinal Obstruction
- ¢ Intestinal Intussusceptions
- ¢ Paralytic ileus





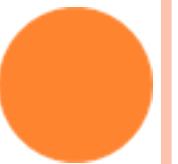


FOREIGN BODY SYNDROME

¢ TRAUMATIC RETICULO PERITONITIS

¢ TRAUMATIC PERICARDITIS







PERITONITIS

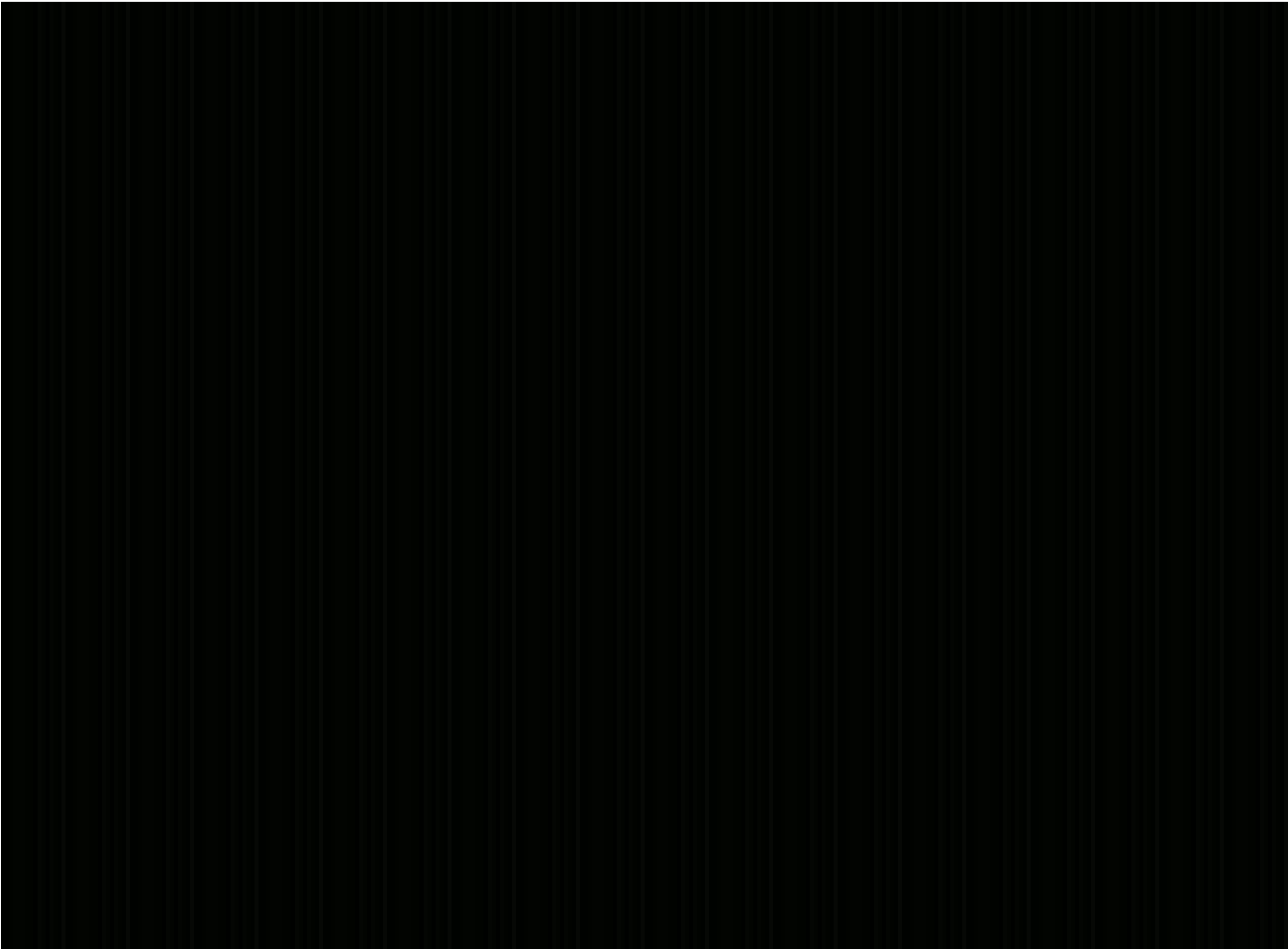
- ¢ Tachycardia
- ¢ Pellety dung
- ¢ Not passing dung
- ¢ Arched back
- ¢ Bruxism
- ¢ Sudden reduction in milk yield
- ¢ Undigested fiber material in rumen fluid and dung

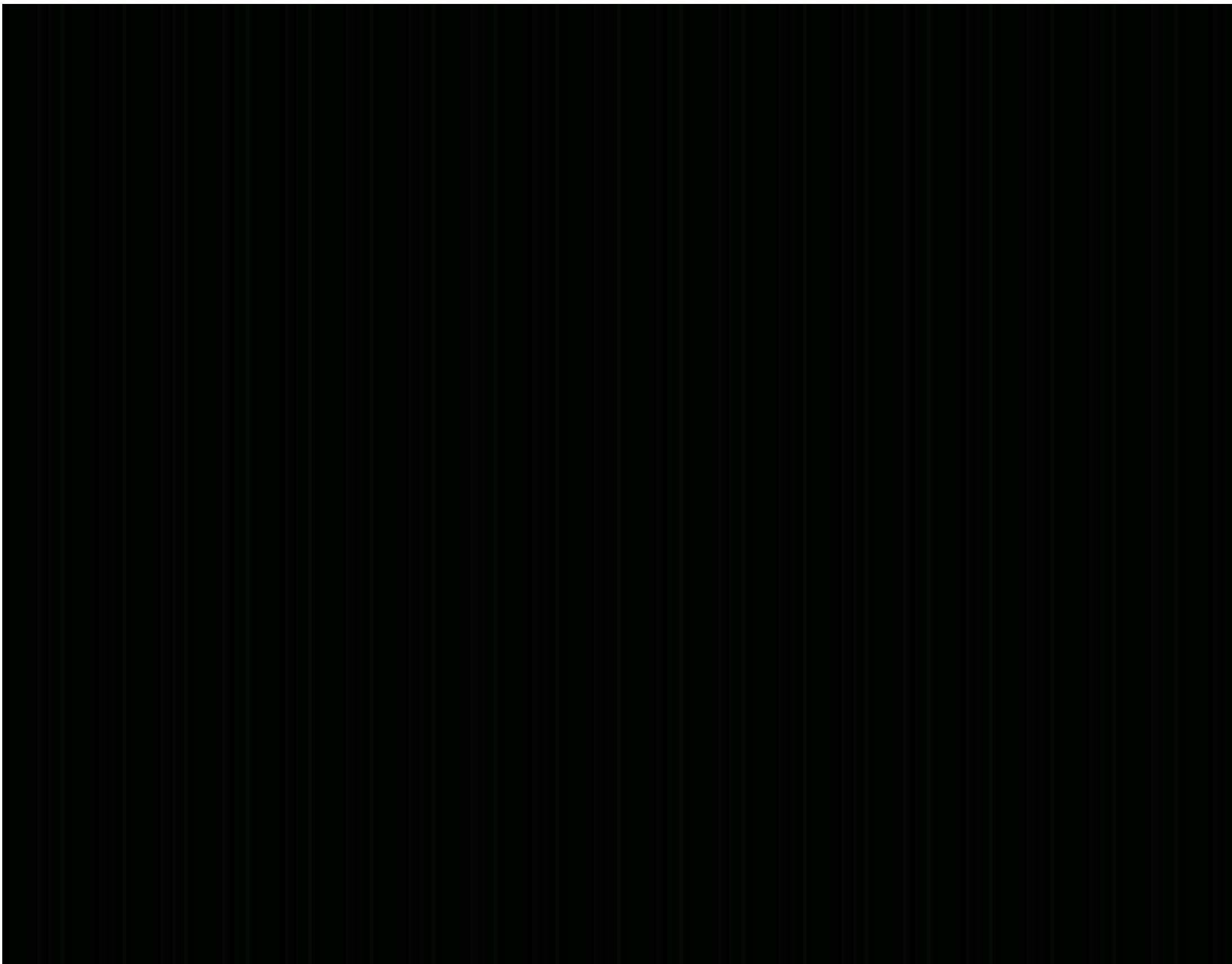




Abdominocentesis









Traumatic Pericarditis



Pericarditis



Venous Stasis Test

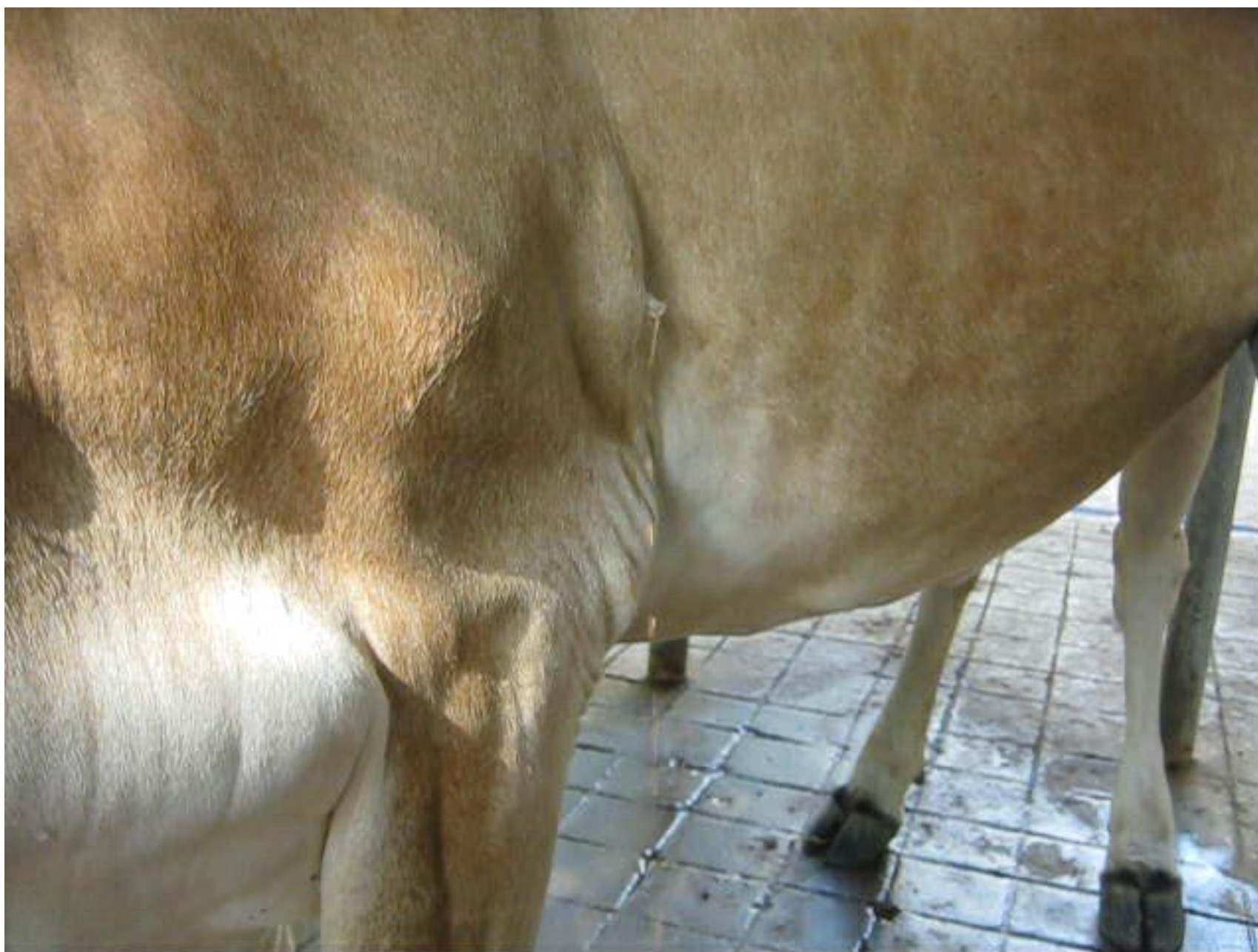


PERICARDITIS

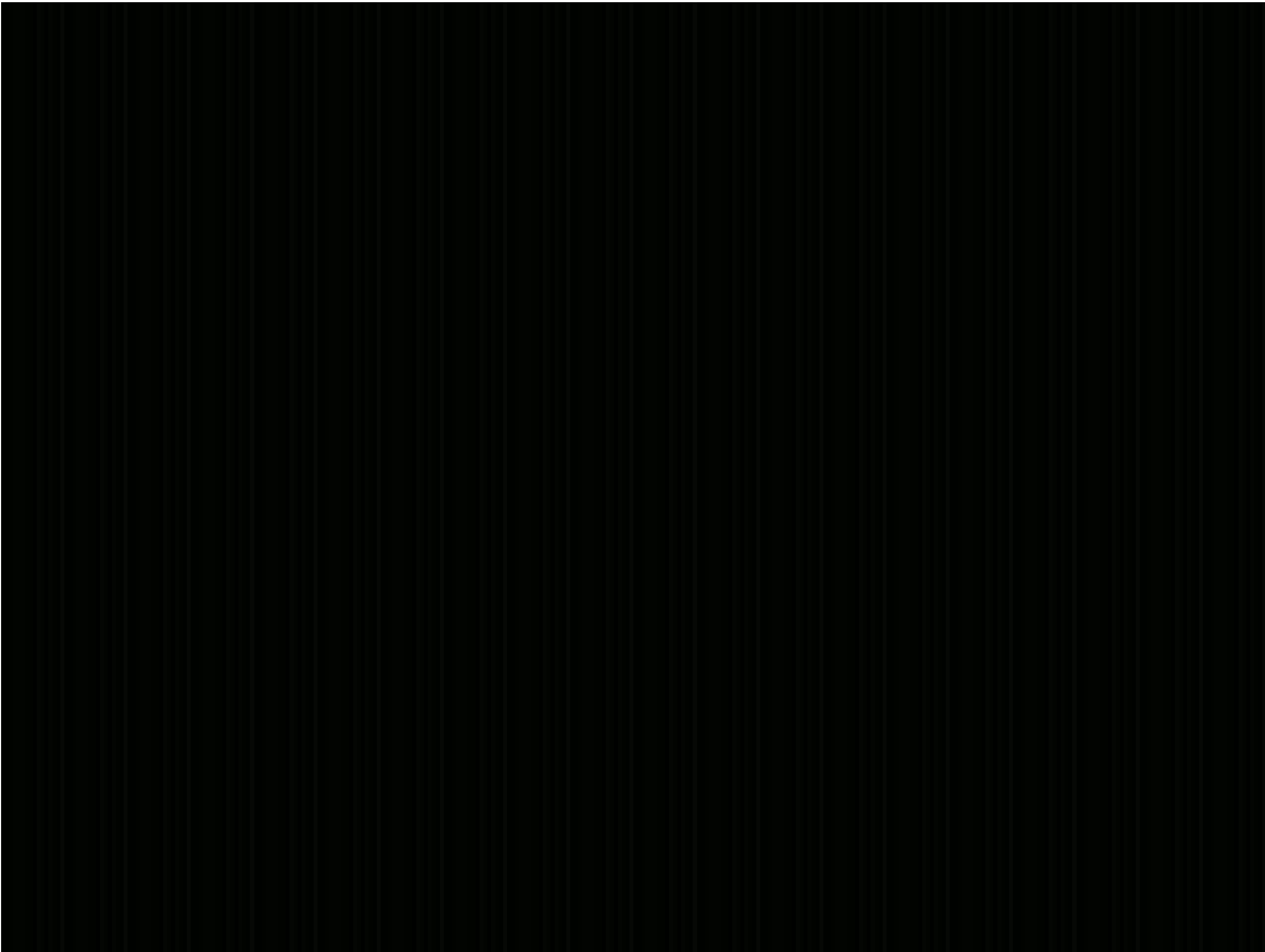
- ¢ Positive venous stasis test
- ¢ Bilateral jugular engorgement
- ¢ Brisket edema
- ¢ Tachycardia
- ¢ Abduction of forelimbs
- ¢ Abnormal Heart sounds on auscultation

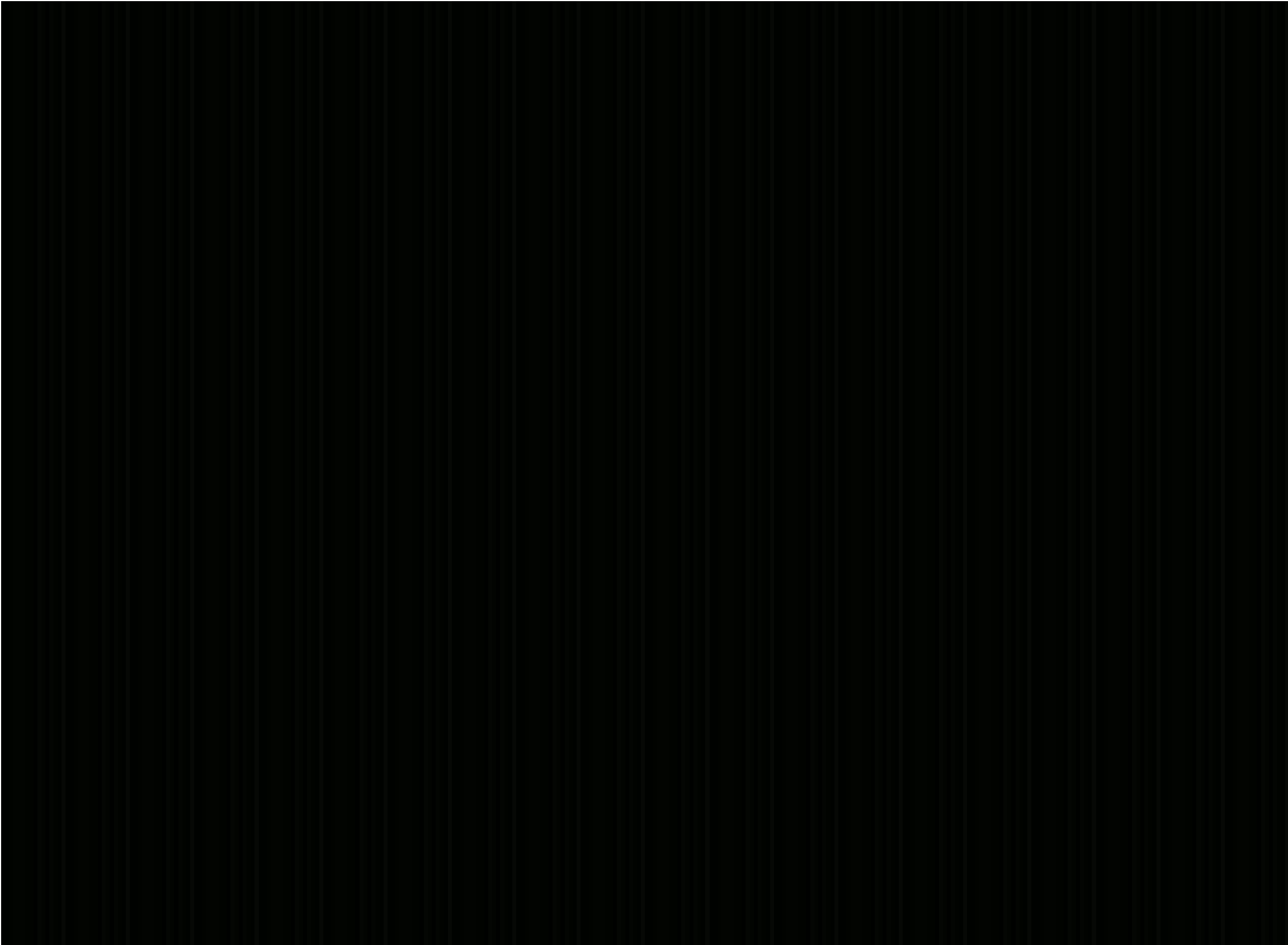












ENTERITIS - DIARHOEA

- ¢ SARA
- ¢ RUMINAL ACIDOSIS
- ¢ BACTERIAL
- ¢ HEMORAGHIC
- ¢ PARASITIC
- ¢ DIPHTHERETIC
- ¢ JD
- ¢ CHF
- ¢ COPPER DEFICIENCY

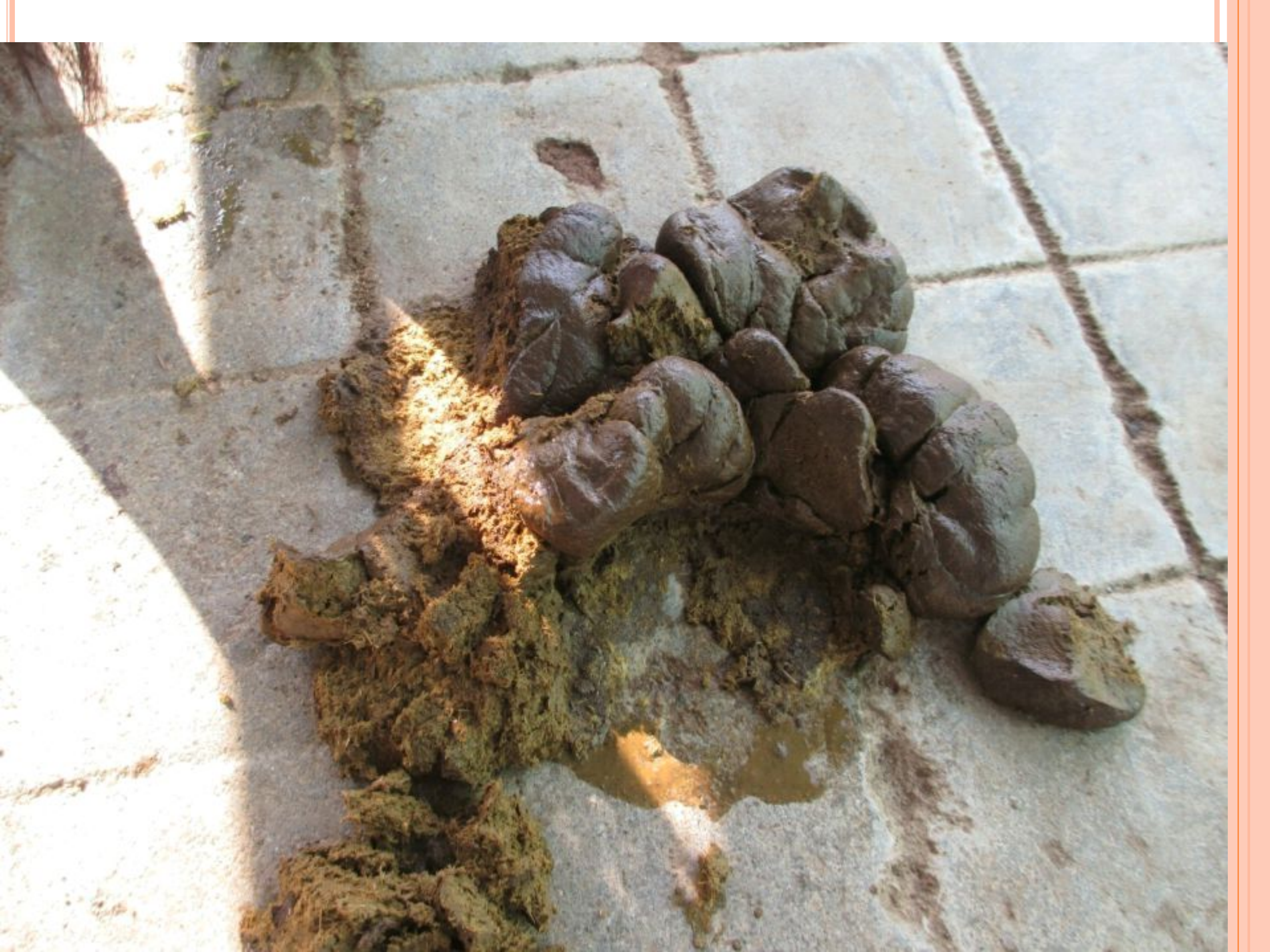
















If TEMPERATURE.....???

- ¢ Ephemeral fever/ Viral
- ¢ Blood Protozoan Diseases
- ¢ Bacterial
- ¢ Inflammatory / Foreign Body
- ¢ Metritis
- ¢ Mastitis



If Icterus...???

- ¢ Anaplasmosis
- ¢ Theileriosis
- ¢ Babesiosis
- ¢ Leptospirosis
- ¢ Photosensitization
- ¢ Blue green algae



If RED COLOURED URINE....

- ¢ Babesiosis
- ¢ Post parturient Hbnuria
- ¢ Bacillary Hbnuria
- ¢ Lepto
- ¢ Phosphorus deficiency
- ¢ Plant poisoning..
- ¢ Miscellaneous?????



If Eye Discharge...???

- ¢ Injury
- ¢ Pink Eye
- ¢ Eye worm
- ¢ Bovine Malignant catarrh
- ¢ Photosensitization



NOT PASSING DUNG/ REDUCED QUANTITY

- ¢ Rumen Impaction
- ¢ Intestinal obstruction
- ¢ Intussusception
- ¢ Paralytic ileus
- ¢ Peritonitis



SALIVATION??

- ¢ Pyrexia
- ¢ Oral Lesions
- ¢ Choke
- ¢ Rabies
- ¢ Intoxication--opc



BLOAT

- ¢ Free gas bloat- choke, tetanus, Frothy bloat
- ¢ Recurrent bloat- DH
- ¢ Tetanus
- ¢ Foreign body (Non penetrating)
- ¢ Postural bloat
- ¢ Cardia obstruction -amphistomes



MUCOUS MEMBRANE COLOUR

- ¢ Pale pink and moist- Normal
- ¢ Pale – anemia, BPD
- ¢ Icteric- anaplasma, theileriosis, Lepto
- ¢ Congested- Toxemia, Septicemia
- ¢ Brick red- HCN
- ¢ Brown- Nitrate poison
- ¢ Blueish/cyanotic- Hypoxia



JOWL EDEMA

- ¢ Parasitic
- ¢ Hypoprotehenemia
- ¢ Hemorrhagic septicemia
- ¢ Actinomycosis
- ¢ Traumatic Pericarditis
- ¢ Snake Bite



PERIPARTURIENT DISEASES

- ¢ Metritis
- ¢ Mastitis
- ¢ Acidosis
- ¢ Peritonitis
- ¢ Pericarditis
- ¢ Hypocalcaemia
- ¢ Hypokalemia
- ¢ Ketosis











HYPOPROTENEMIA - BUFFALO



DERMATOPHILOSIS







CHEMOSIS



Discussions?????????



THANK YOU

